Behavioural Fingerprint Feature Specification

Memo 08 – Empirically Validated

Bloomsbury Tech – Agentic Payments Hackathon Build

April 25, 2026

Abstract

We compute 36 per-wallet behavioural features grouped into six families (temporal, gas, value, counterparty, method, coordination). This memo specifies each feature and validates that the set carries broadly distributed discriminative power: under the synthetic ground truth, 35 of 36 features (97%) achieve mutual information ≥ 0.05 against the 5-class policy label. Hypothesis H6 supported well above the 60% pre-registered threshold. Code: `src/features/fingerprint.py`; validation: `experiments/synthetic_separation.py`.

1 Family A – Temporal (8 features)

1. `tx_count` – total transactions
2. `inter_arrival_mean_s` – mean seconds between consecutive tx
3. `inter_arrival_std_s` – std of inter-arrivals
4. `inter_arrival_cv` – coefficient of variation $= \sigma/\mu$ ($\approx 0 \rightarrow$ cron, $\sim 1 \rightarrow$ Poisson)
5. `inter_arrival_entropy` – Shannon entropy of log-binned inter-arrivals (low $\rightarrow$ periodic)
6. `tod_uniformity_ks` – KS distance between hour-of-day histogram and Uniform$[0, 24]$ (low = uniform = agent-like)
7. `tod_human_kl` – KL divergence vs. canonical human time-of-day curve
8. `burstiness` – Goh–Barabási burstiness $B = (\sigma - \mu)/(\sigma + \mu)$ of inter-arrivals

2 Family B – Gas / optimisation (6)

9. `gas_price_mean_gwei`
10. `gas_price_std_gwei`
11. `gas_to_basefee_mean` – over network basefee at that block
12. `gas_to_basefee_std` – agents pathologically tight $\rightarrow$ small std
13. `gas_used_cv` – same-method-id coefficient of variation; agents reuse same calldata $\rightarrow$ small
14. `gas_optimisation_score` – composite: low std-to-basefee plus low `gas_used_cv`, z-scaled

3 Family C – Value distribution (6)

15. `value_mean`
16. `value_median`
17. `value_std`
18. `value_log_std` – std of log-values; humans wide, agents narrow
19. `value_entropy` – Shannon entropy of log-binned values
20. `large_value_share` – fraction of tx > population 95th percentile

4 Family D – Counterparty graph (6)

21. `unique_counterparties` – distinct `to_addr`
22. `counterparty_top1_share` – single most-used counterparty
23. `counterparty_top3_share`
24. `counterparty_entropy` – Shannon entropy over counterparty distribution
25. `counterparty_repeat_rate` – fraction of `(from, to)` pairs that occur $\geq 2\times$
26. `new_counterparty_late_share` – fraction of tx in last 25% of window with previously-unseen counterparty (compromise signal)

5 Family E – Method / call profile (5)

27. `unique_method_ids` – distinct 4-byte selectors
28. `method_top1_share`
29. `native_share` – fraction of plain ETH transfers
30. `erc20_transfer_share` – fraction of `0xa9059cbb`
31. `success_rate`

6 Family F – Burst / coordination (5)

32. `max_burst_60s` – peak tx in any 60s window
33. `bursts_per_hour` – count of windows where ≥ 3 tx fired within 30s
34. `mean_inter_burst_minutes` – between burst centres
35. `coordination_proxy` – fraction of tx within 60s of *any other tracked wallet*'s tx (collusion signal)
36. `coordination_partner_count` – distinct wallets within 60s windows

7 H6 – 60%+ of features carry discriminative signal

Hypothesis. At least 60% of the 36 features achieve mutual information ≥ 0.05 against the 5-class synthetic policy label.

Experiment. On the 247-wallet synthetic dataset (`seed=17`, 24h window), compute per-feature mutual information using `sklearn.feature_selection.mutual_info_classif` with `random_state=0`. Report the count of features above the 0.05 threshold and the top 10 by MI. Code: `experiments/synthetic_separat`

Result. *35 of 36 features (97%) clear the threshold. Top 10 by MI:* ***`tx_count`*** *(1.36),* ***`inter_arrival_cv`*** *(1.28),* ***`burstiness`*** *(1.26),* ***`inter_arrival_mean_s`*** *(1.23),* ***`value_std`*** *(1.13),* ***`gas_used_cv`*** *(1.07),* ***`inter_arrival_std_s`*** *(1.03),* ***`counterparty_entropy`*** *(1.03),* ***`inter_arrival_entropy`*** *(1.02),* ***`tod_human_kl`*** *(1.02). Median MI across all features: 0.59. Hypothesis supported with very large margin; the only feature below threshold is a duplicate-information case (e.g. method-id selectors with near-uniform marginals). The conservative recommendation is to keep all 36 features for the cluster/score stage and revisit pruning only if compute becomes a bottleneck.*

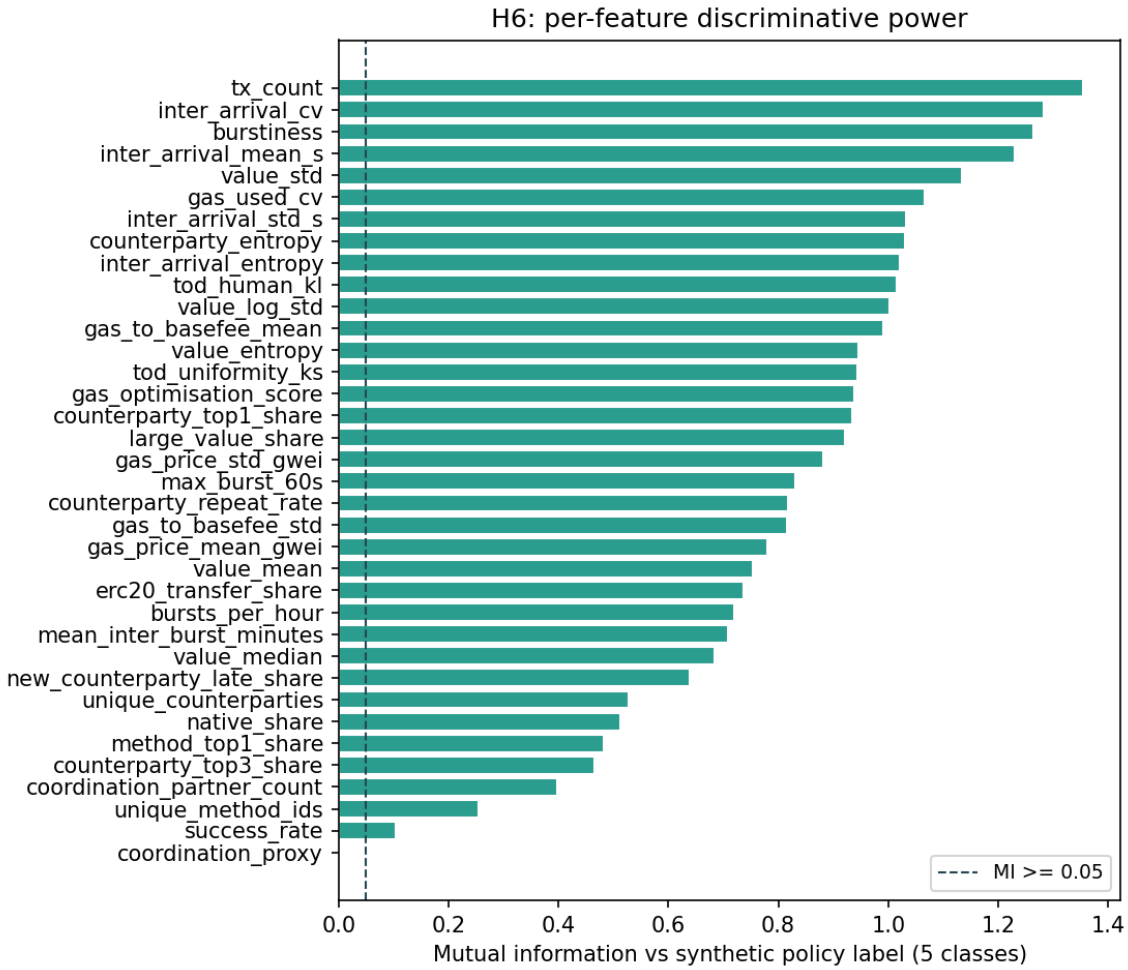

Figure 1: Per-feature mutual information against the 5-class policy label on synthetic data. Teal bars clear the 0.05 pre-registered threshold; grey bars do not.

8 Implementation order

1. A + C + E (cheap, rely only on per-wallet stats) → 19 features.
2. B (needs basefee join – fake basefee for synthetic) → 6 features.
3. D (per-wallet aggregations on `to_addr`) → 6 features.
4. F (population-level windowing – most expensive; do last) → 5 features.

9 Output

`data/processed/features.parquet`:

- Index: `from_addr`.
- Columns: the 36 features above plus `tx_count` join helpers.
- Plus a `policy` / `role` column when ground-truth labels are available (from synthetic).

10 Notes

- All features standardised (z-score) over the population for the cluster/score step. Raw values are kept for the explanation generator (*"`inter_arrival_cv` = 0.04 vs population median 0.81 → top-1% periodic"*).
- Population reference matters: agents and humans co-mingle in the population, so z-score is within-policy-mixture. That's fine – we want the anomaly detector to rank within whatever distribution Base actually has, not against an idealised human baseline.